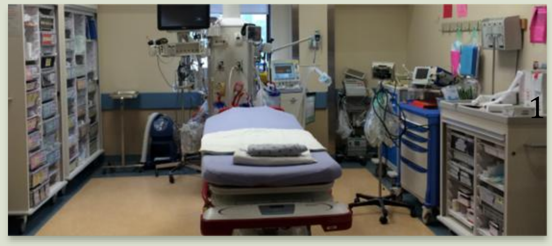
Multi-trauma: Burn + Head Injury



Section I: Scenario Demographics

Scenario Title:	Multi-trauma Burn + Head Injury
Date of Development:	29/04/2016 (DD/MM/YYYY)
Target Learning Group:	☐ Juniors (PGY 1 – 2) ☒ Seniors (PGY ≥ 3) ☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Donika Orlich (adapted from "Two Patient Trauma" by Martin Kuuskne)
Affiliations/Institution(s):	McMaster University
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To expose learners to a dual patient case which includes a severe burn patient requiring escharotomy.
CRM Objectives:	1) Demonstrates resource utilization by calling for help appropriately, delegating tasks when needed and triaging patients appropriately 2) Communicates clearly with an inter-professional team during handover over patient care and during patient management
Medical Objectives:	1) Initiates appropriate work-up and treatment for suspected CO/CN poisoning 2) Recognizes the potential for a difficult intubation and plans accordingly 3) Considers escharotomy in a difficult to ventilate burn patient and performs the procedure 4) Recognizes hypoglycemia in the altered patient with HI and initiates treatment

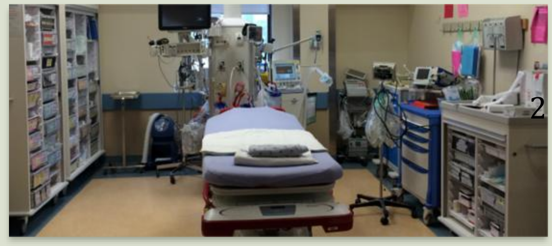
Case Summary: Brief Summary of Case Progression and Major Events

The case will begin with the arrival of patient from a house fire who has 30%TBSA burns. The team will be expected to recognize the need for intubation and fluid resuscitation. After successful intubation, a second patient will arrive from an altercation outside a bar. He appears to have a blunt traumatic head injury after being repeatedly kicked. The team is expected to recognize hypoglycemia in the context of a minor head injury and provide immediate glucose replacement. During the management of the head injured patient, the burn patient will continue to by hypotensive. The team will need to recognize the possibility of CN toxicity. The patient will also become more difficult to ventilate and will require an escharotomy.

References

Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby.

Website: Trauma.org. Accessed May 2, 2016. *Emergency Department Thoracotomy*. http://www.trauma.org/index.php/main/article/361/



Section IV: Scenario Script

A. Clinical Vignette: To Read Aloud at Beginning of Case

Patient A: "You are working in a tertiary care ED. A 33 year old male has just been brought in by EMS after being dragged out of a house fire. He has been unresponsive with EMS and has significant burns to his chest, arm, and leg. The etiology of the fire is unclear, but the home was severely damaged."

Midway through the case, Patient B will arrive with EMS:
Patient B: "55 year old male, repeatedly kicked during an altercation outside a bar. GCS 15 on arrival, but just decreased to 13 in the ambulance bay, and he has become combative. C-spine collar applied immediately. Lots of bruising to face/head, but no other obvious injuries. Patient denied other medical history or allergies initially".

B. Scenario Cast & Realism

Patients: For both patients	☒ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient	*Select most important dimension(s)*	☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A

Confederates	Brief Description of Role
Paramedic	To give HPI upon transfer of 2nd patient (as scripted above).
Nurse x2	To assist with cues to patient status, medication administration, etc. (One nurse per patient will be required to run case smoothly)

C. Required Monitors

☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line
☒ NIBP Cuff	☐ Defibrillator Pads	☐ Capnography
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:

D. Required Equipment

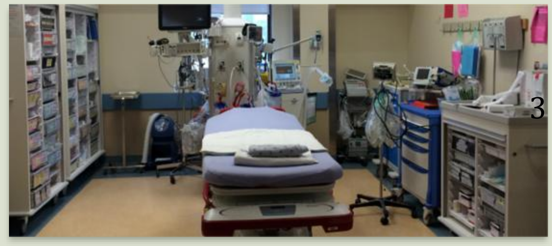
☒ Gloves	☒ Nasal Prongs	☒ **Scalpel**
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit
☐ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit
☒ IV Push Medications	☒ Laryngoscope	☒ **Central Line Kit**
☐ PO Tabs	☐ Video Assisted Laryngoscope	☒ **Arterial Line Kit**
☐ Blood Products	☒ ET Tubes	☐ Other:
☐ Intraosseous Set-up	☐ LMA	☐ Other:

E. Moulage

Patient A: Soot on face, singed eyebrows. Fake burns to chest/back, right arm/abdomen/leg (circumferential to chest). C-collar in situ. (Note: can moulage burns with black garbage bag stuffed with red bubble wrap and tissue paper so that students can "cut" for escharotomy)
Patient B: Bruising/contusions over face and head. Missing front tooth.

F. Approximate Timing

Set-Up:	5 min	Scenario:	20 min	Debriefing:	40 min



Section V: Patient Data and Baseline State- PATIENT A

A. Clinical Vignette: To Read Aloud at Beginning of Case

"You are working in a tertiary care ED. A 33 year old male has just been brought in by EMS after being dragged out of a house fire. He has been unresponsive with EMS and has significant burns to his chest, arm, and leg. The etiology of the fire is unclear, but the home was severely damaged. Last vitals: HR 120, BP 130/80, RR 30, O2 95% NRB. GCS 3 entire time with them, tolerating an oral airway."

B. Patient Profile and History

Patient Name: Shawn Sherman	Age: 33	Weight: 70	
Gender: ☒ M ☐ F	Code Status: Full		
Chief Complaint: burns, aLOC			
History of Presenting Illness: **As above. Pulled from a house fire.**			
Past Medical History:	Unknown	Medications:	Unknown
Allergies: None known.			
Social History: Unknown			
Family History: Unknown.			

Review of Systems:				
	CNS:	Unable		
	HEENT:	Unable		
	CVS:	Unable		
	RESP:	Unable		
	GI:	Unable		
	GU:	Unable		
	MSK:	Unable	INT:	Unable

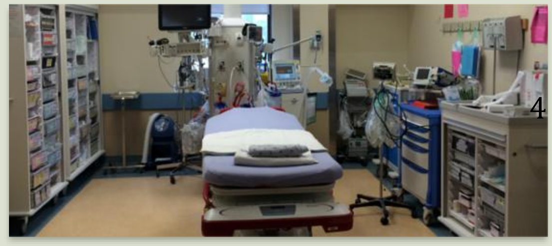
C. Baseline Simulator State and Physical Exam

☐ No Monitor Display ☒ Monitor On, no data displayed ☐ Monitor on Standard Display

HR: 130/min	BP: 90/70	RR: 30/min	O_2SAT: 95% NRB
Rhythm: sinus tach with multiple PVCs	T: 36.1°C	Glucose: 6.1 mmol/L	GCS: 3 (E 1 V 1 M1)

General Status: Unresponsive.

CNS:	**GCS 3. Pupils 3mm bilat, minimally reactive.**		
HEENT:	No signs HI. **Soot on face. Singed eyebrows.**		
CVS:	Nil.		
RESP:	GAEB. No adventitious.		
ABDO:	Abdo soft, NT.		
GU:	Nil.		
MSK:	No signs trauma. Pelvis stable.	SKIN:	**Burns to entire chest/back, R arm, R leg.**

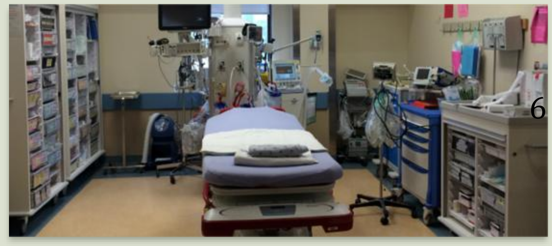


Section VI: Scenario Progression –Patient A

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: Sinus tach + frequent PVCs HR: 130/min BP: 90/70 RR: 30/min O_2SAT: 96 % NRB T: 36.1°C	Unresponsive with GCS 3.	<u>Learner Actions</u> -☐ 2 large bore IVs, bolus 2L -☐ 100% O2, monitors -☐ Labs: VBG, carboxyHb, lactate, coags, trop, G&S, INR -☐ Check **glucose: 6.3** -☐ Portable CXR/PXR -☐ Full exposure re. TBSA -☐ US FAST exam	<u>Modifiers</u> *Changes to patient condition based on learner action* - 2L IVF → HR 120, BP 120/80 <u>Triggers</u> *For progression to next state* - Intubate → **2. Intubation** - 7 minutes → **3. Critical VBG**
2. Intubation HR: 120 BP: 90/70 RR: 30/min O_2SAT: 93 % NRB	Unchanged.	<u>Learner Actions</u> -☐ Push dose pressors at bedside -☐ Consider apneic oxygenation -☐ Difficult airway cart -☐ Surgical airway kit at bedside -☐ Consider 1st look before paralytic -☐ Anesthesia consult	<u>Modifiers</u> - NE started → BP 95/75 - Propofol used → BP 60/30 (BP 80/50 if NE started) - Any other agent used → BP 80/60 <u>Triggers</u> **-** Intubation complete → **3. Critical VBG** ****PATIENT B ARRIVES**
3. Critical VBG HR: 100 BP: 80/60 RR: 12/min vented O_2SAT: 95%	Unchanged.	<u>Learner Actions</u> -☐ Ensure patient on 100% O2 -☐ Continue IVF to replace 3rd space loses -☐ Start vasopressor if not already done -☐ Hydroxycobalamin 5mg IV -☐ ± Na thiosulfate 12.5g IV -☐ ± Call Poison Centre	<u>Modifiers</u> - No hydroxycobalamin given 5 min into state → **RN to prompt** "isn't there some medicine you give when people are in a fire?" <u>Triggers</u> - Hydroxycobalamin given → **4. Ventilator Alarming**
4. Ventilator Alarming HR: 110 BP: 90/50 RR: 12/min vented O_2SAT: 85%	"High pressure" alarms on vent and difficult to vent or bag. Nurse to help prompt re: vent alarms for high pressure.	<u>Learner Actions</u> -☐ Disconnect vent and attempt manual BVM -☐ Consider US for PTX or needle decompression -☐ Manual decompression of chest -☐ Escharotomy	<u>Modifiers</u> - Manual decompression → no change - Escharotomy → "improved ease of ventilation", O_2 95% <u>Triggers</u> - Escharotomy → **5. Resolution** - No escharotomy by 20mins → **END CASE**
5. Resolution HR: 110 BP: 100/50 RR: 12/min vented O_2SAT: 85%	Patient remains unresponsive.	<u>Learner Actions</u> -☐ Call ICU/Plastics -☐ Call Poison Centre -☐ ± Call for hyperbarics	<u>ICU arrives to manage patient</u> **END CASE**

Section VII: Supporting Documents, Laboratory Results, & Multimedia – Patient A

Laboratory Results					
VBG	pH: 6.98	PCO_2: 28	PO_2: 40	HCO_3: 6	Lactate: 11
Carboxyhgb: 0.4					



Section VIII: Patient Data and Baseline State - PATIENT B

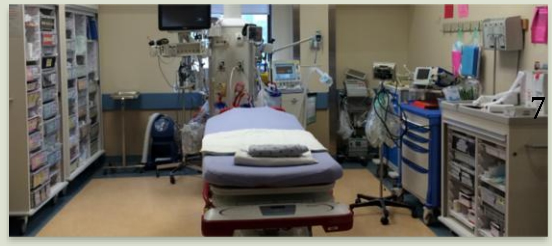
A. Patient Profile and History		
Patient Name: Michael Bissidy	Age: 55	Weight: 150kg
Gender: ☐ M ☒ F	Code Status: Unknown	
Chief Complaint: Head injury		
History of Presenting Illness: Patient involved in altercation outside a bar. Bystanders report the patient was repeatedly kicked in the head/face after stabbing another person. Initially GCS 15 with EMS, but decreased to GCS 13 as they pulled up to the hospital and became ++ agitated pulling out IV.		

Past Medical History:		Medications:	
	Hypertension		Hydrochlorothiazide
	Dyslipidemia		Atorvastatin
	Diabetes		Metformin
			Insulin

Allergies: nil				
Social History: unknown				
Review of Systems:	CNS:	Initially complaining of headache to EMS. Now unable.		
	HEENT:	**"Face hurts"**		
	CVS:	No complaints.		
	RESP:	No complaints.		
	GI:	No complaints.		
	GU:	No complaints.		
	MSK:	No complaints.	INT:	No complaints.

B. Baseline Simulator State and Physical Exam			
☐ No Monitor Display	☒ Monitor On, no data displayed	☐ Monitor on Standard Display	
HR: 90 /min	BP: 120/80	RR: 20/min	O_2SAT: 99% RA
Rhythm: sinus	T: 35.6°C	Glucose: **2.6 mmol/L**	GCS: 13 (E4 V4 M5)
General Status: **Very large muscular/obese man. Combative. Swearing**			

CNS:	**Seems confused. ++ Agitated.**		
HEENT:	**Significant bruising over left orbit/jaw, bleeding nose, laceration to scalp with hematoma (bleeding controlled)**		
CVS:	Palpable, strong, tachycardic, pulse in all extremities. No murmurs. Normal heart sounds		
RESP:	GAEB, no adventitious sounds		
ABDO:	Normal		
GU:	Normal		
MSK:	No extremity deformities	SKIN:	**Bruising/laceration to head & face**

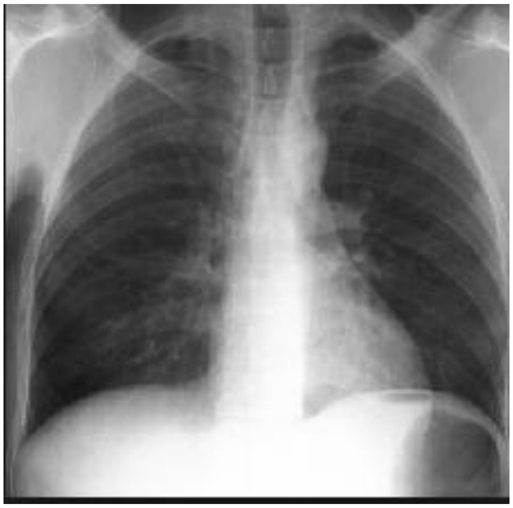


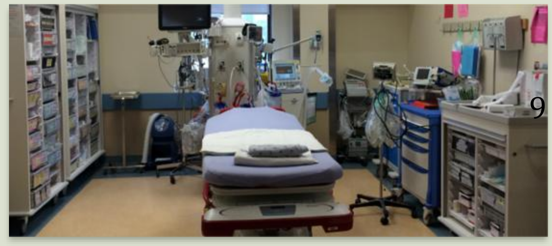
Section IX: Scenario Progression –Patient B

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: NSR HR: 110/min BP: 120/78 RR: 20/min O_2SAT: 99 % RA T: 35.6^{o}C	GCS – 13 Seems confused. Patient is very agitated, swearing and trying to get off EMS stretcher and pull off his c-spine collar. Patient is collared.	Learner Actions -☐ Call for security/back-up -☐ IM sedation -☐ Monitor, full vitals -☐ Establish IV access -☐ Take history from EMS -☐ Perform primary survey -☐ Check sugar: **2.6 mmol/L** -☐ Replace glucose with D50 -☐ Send trauma labs -☐ ± eFAST -☐ Progress to secondary survey -☐ Determine need for further imaging (CT head, C-spine films)	Modifiers *Changes to patient condition based on learner action* - IM sedation -> settles over 1 min - If eFAST done → negative - If team decides needs imaging → radiology will ask who to image first Triggers *For progression to next state* - If cap sugar not checked by 5 min → **2. Seizure** - If sugar replaced, primary survey complete → **3. Normal GCS**
2. Seizure HR: 145 BP: 140/75 RR: 12	Patient has tonic/clonic seizure.	Learner Actions -☐ Check sugar: **2.6 mmol/L** -☐ Replace glucose with D50 -☐ Arrange for CT head -☐ Consider other causes of seizure/aLOC	Modifiers - If preparing to intubate, RN to ask: "should we check a sugar first?" Triggers - Glucose replaced → **3. Normal GCS**
3. Normal GCS HR: 90/min BP: 110/78 RR: 18/min O_2SAT: 99 %	GCS 15 Patient alert and oriented	Learner Actions -☐ Send trauma labs -☐ ± eFAST -☐ Progress to secondary survey -☐ Determine need for further imaging (CT head, C-spine films)	**END CASE PRN**

Section X: Supporting Documents, Laboratory Results, & Multimedia – Patient B

Laboratory Results - Patient B
Not available

Images (ECGs, CXRs, etc.)	
Patient B – CXR  CXR source: http://www.pharmacology2000.com/respiratory_anesthesiology/pulmonary_assessment/pulmonary_assessment2.htm	Patient B – Normal Pelvic Xray  PXR source: http://radiopaedia.org/articles/pelvis-1



Section XI: Debriefing Guide

General Debriefing Plan			
☐ Individual	☒ Group	☐ With Video	☒ Without Video

Objectives	
Educational Goal:	To expose learners to a dual patient case which includes a severe burn patient requiring escharotomy.
CRM Objectives:	1) Demonstrates resource utilization by calling for help appropriately, delegating tasks when needed and triaging patients appropriately 2) Communicates clearly with an inter-professional team and taking and giving effective handover
Medical Objectives:	1) Initiates appropriate work-up and treatment for suspected CO/CN poisoning 2) Recognizes the potential for a difficult intubation and plans accordingly 3) Considers escharotomy in a difficult to ventilate burn patient and performs the procedure 4) Recognizes hypoglycemia in the altered patient with HI and initiates treatment

Sample Questions for Debriefing
1. How was the decision made to split the team and resources? Do you think it was done well? 2. What was the team leader's leadership style? Did it change when the second patient arrived? 3. What difficulties are faced when using resources for a trauma involving multiple patients? 4. What are the indications for an escharotomy?

Key Moments
1) Recognition that the team must be split to appropriately manage both patients.
2) Recognition of hypoglycemia as the cause for aLOC in patient B
3) Identification of CN as cause for persistent hypotension
4) Recognition of circumferential burns as the cause for high ventilator pressures
5) Decision to perform escharotomy